



FPGA-Based Bank Queue Management System

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Introduction to Queue Management Systems

Addressing Inefficiencies

Traditional bank queues suffer from significant inefficiencies. This leads to long wait times and frustrated customers.

Customer Dissatisfaction

Studies reveal customers wait over 20 minutes on average. This impacts their overall banking experience negatively.

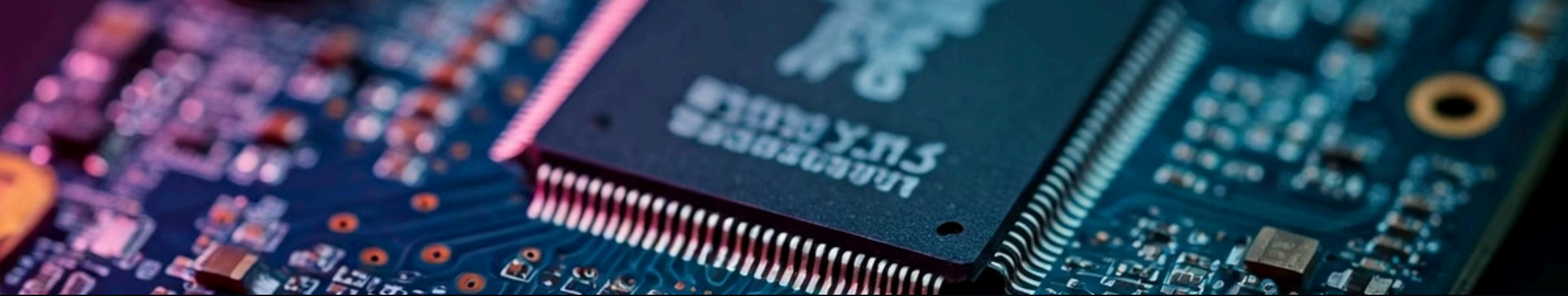
Modern Solution Needed

There's a critical need for modern, efficient systems. Our solution aims to revolutionize this experience.



Abstract

This project presents a hardware-based Queue Management System implemented on the Digilent Basys 3 FPGA development board using Verilog HDL. Designed to simulate a real-world environment such as a bank or service center, the system manages a queue of customers and distributes them between two service counters (tellers) efficiently. The system utilizes onboard push buttons, slide switches, LEDs, and seven-segment displays to allow customer interaction, teller operation, and real-time display of queue status. Key features include passcode-protected administrative control, dynamic counter allocation, and clear status indication for open/closed service counters. The project emphasizes digital design principles, state machine control, and real-time user interface implementation on FPGA hardware. Development and testing were conducted using the Xilinx Vivado Design Suite.



Why Use an FPGA?

High Performance

FPGAs offer real-time processing and superior performance. They can be 10x faster than microprocessors.

Parallel Processing

They enable parallel operations for multiple queues. This ensures simultaneous handling of tasks.

Real-Time Control

FPGAs give **instant response** to button and switch presses. This is perfect for a **live queue system**, where delay is not acceptable.

Custom Logic

You can write **your own passcode rules**, counter open/close logic, and customer call order — all in **Verilog**. No fixed software needed.

Components & Softwares



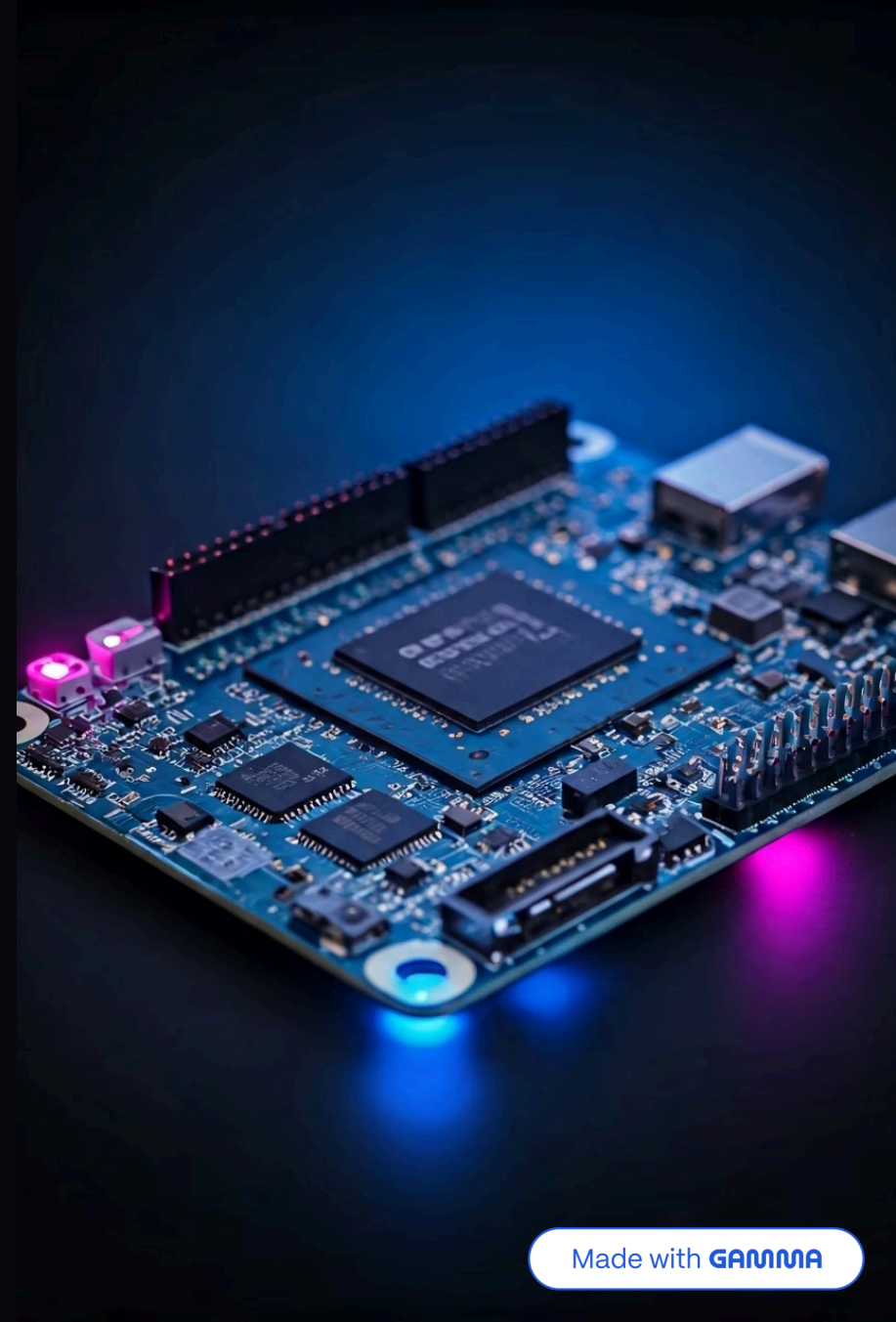
Hardware Components:

- **FPGA Board:** Digilent Basys 3 (Artix-7 FPGA)



Software Tools:

- **Xilinx Vivado Design Suite** (for coding, simulation, synthesis, and programming)
- **Verilog HDL** (hardware description language used)



Working Principle

1

This project is developed on the Basys 3 FPGA board using Verilog and mimics a bank queue system where customers press a button to join the line, and tellers press buttons to call the next customer.

2

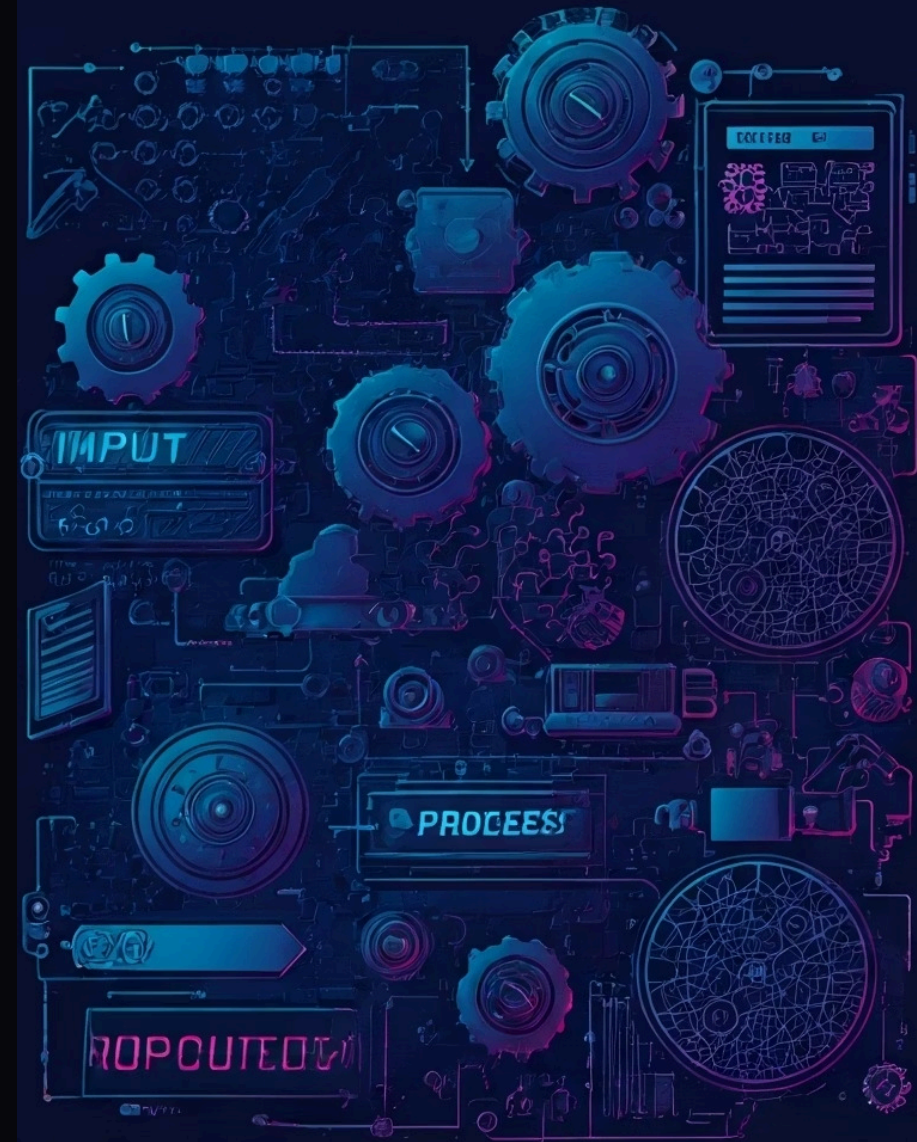
LEDs and 7-segment displays indicate whether service counters are open, show the last served customer number, and display the number of people waiting.

3

A secret passcode entered using switches allows the system to open or close the bank for added security.

4

Debouncing modules are used to remove button noise, and display control modules handle output, with the system tested through simulation before running on the actual board.



Inputs



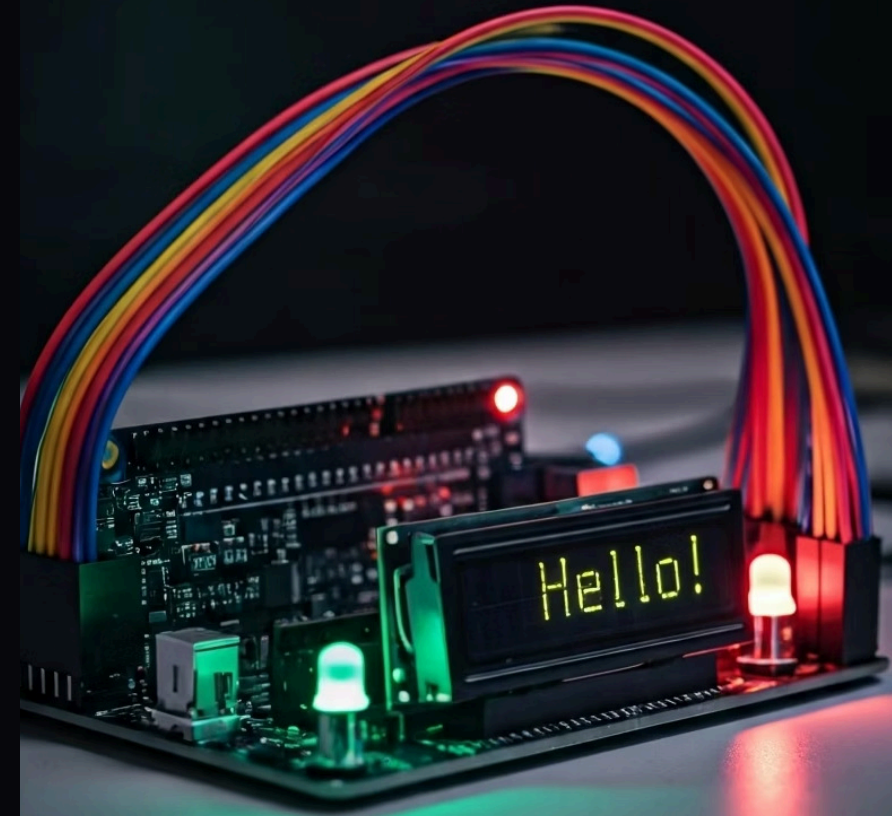
- Passcode switches (Rightmost 12 switches): Used to enter passcode
- Bank switch (Leftmost switch): Used to open/close bank
- Counter switch (2nd leftmost switch): Used to open/close counter
- Passcode change switch (3rd leftmost switch): Used to change passcode
- Reset button (Lower button): Used to reset system
- Customer button (Upper button): Used for taking queue number
- Teller buttons (Right and left button): Used for teller actions, effects to the system changes according to the leftmost three switches.



Outputs



- Bank LED (Leftmost LED): Shows if the bank is open or not
- Counter LEDs (Rightmost two LEDs): Shows if the corresponding counter is open or not
- Wait LED (9th LED from left site): Shows if the system is waiting for teller action to complete a process
- Seven Segment Displays: Can show different things according to other signals
 - Default: Last called customer for the tellers (Right two for one teller, left two for the other)
 - When customer takes number: For one, second left two seven-segment displays are blank, right two seven-segment displays show taken number. Seven-segment displays blink within a period of 0.5 second and duty cycle of 50%.
 - Bank closed: Writes "CLSd"
 - If tellers try to close bank while there is/are customer(s) waiting and/or there is an open counter: "rCXX", where XX number of remaining customer(s)



Advantages and Disadvantages

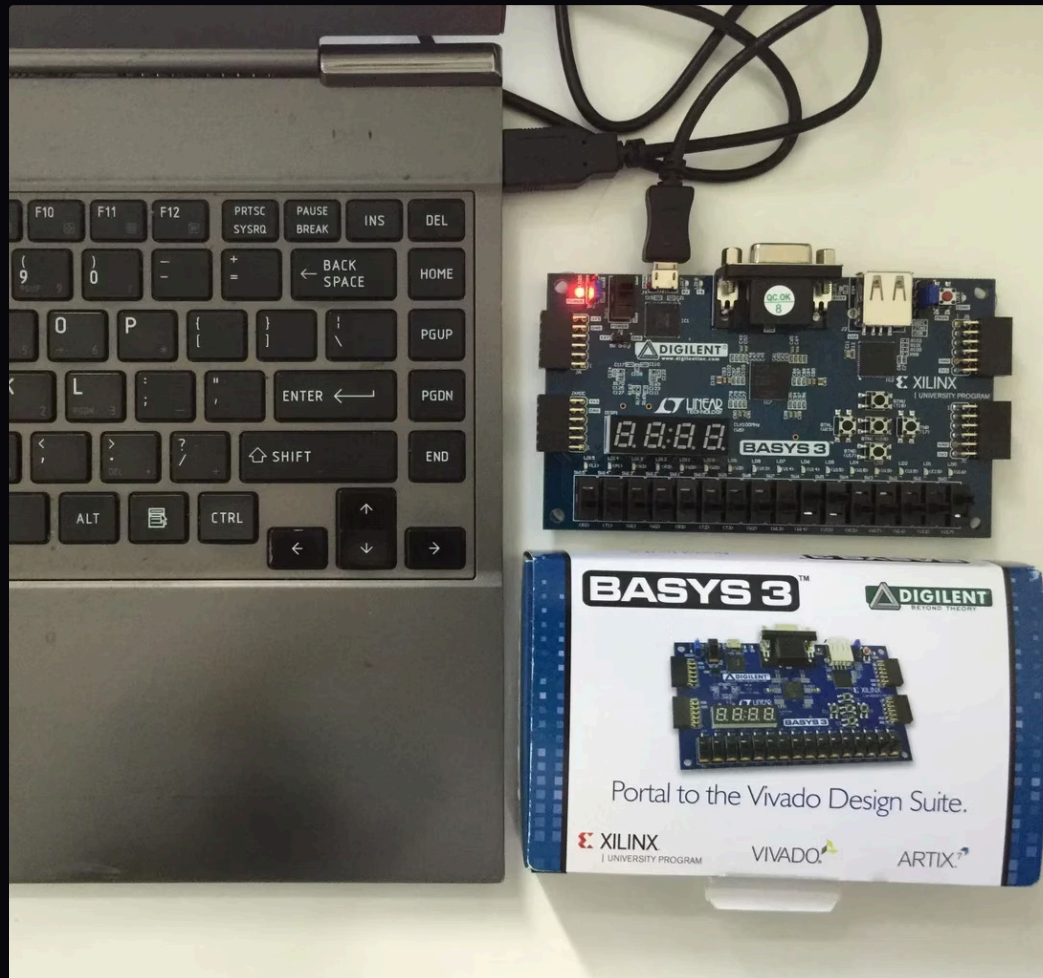
Advantages

- Efficient queue handling
- Secure access with passcodes
- Real-time fast operation using FPGA
- Simple user interface with buttons and LEDs

Disadvantages

- Limited to only 2 tellers and fixed customer range
- Passcodes are entered manually via switches
- No remote access or monitoring
- Doesn't store full customer queue history

Circuit Diagram



Results

Case-1 : When No Customer is there P

Results

Case-2 : When a Customer is there and we are trying to close the bank

Applications

📌 Banking Sector

Manages customer flow efficiently by assigning tokens and reducing queue confusion.

📌 Hospitals & Clinics

Helps doctors and staff call patients in an organized manner, improving service time.

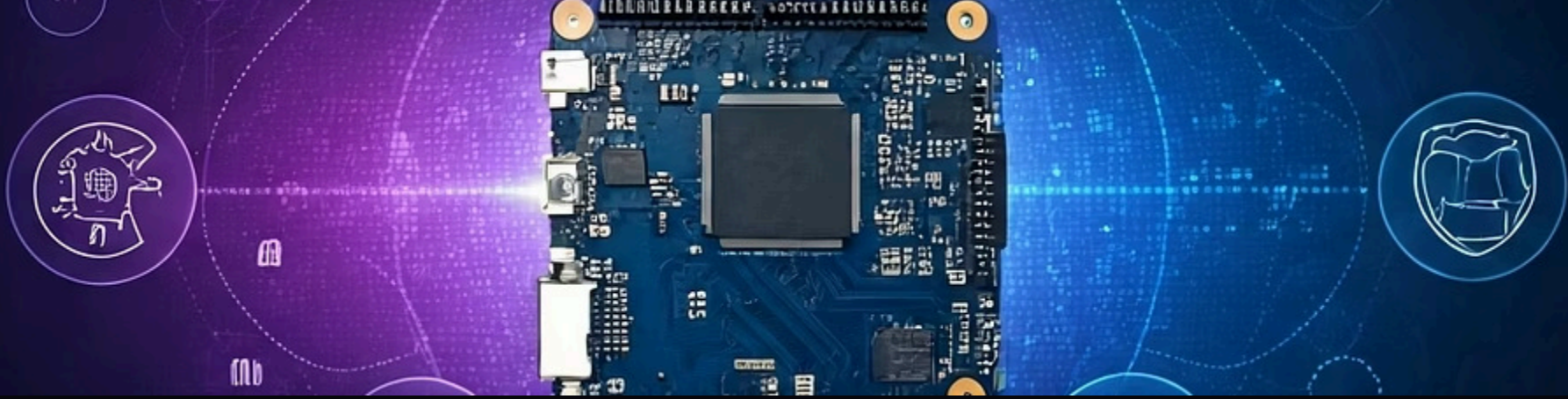
📌 Government Offices

Useful in places like passport offices or RTOs where people wait for their turn.

📌 Customer Service Centers

Applies to billing counters, help desks, or customer support areas to manage crowds.





Conclusion

This project demonstrates a simple and effective queue management system using the Digilent Basys 3 FPGA board. It allows secure teller login through passcodes, organized customer token handling, and proper control of bank and counter operations using buttons and switches. The system ensures smooth coordination between customers and tellers in real time. Overall, it highlights the practical use of FPGA technology in building smart and reliable service systems like those used in banks, hospitals, and other public offices.

Thank You

